

Profit Function Properties: Summary

In what follows let $Y \subset \mathbb{R}^L$ be a nonempty, closed production set. Consider

$$\pi(p) = \max_{y \in Y} p \cdot y \quad (1)$$

and let P be the (convex) set of prices for which $\pi(p)$ is finite. Of course problem (1) is mathematically identical to $-\pi(p) = \min_{y \in Y} p \cdot (-y)$, which in turn is *almost* identical in structure to the expenditure minimization problem. (Since a production set is not generally bounded above, $-Y$ is not bounded from below, unlike the consumption set X .) The next Corollary therefore follows from our proof of the expenditure function properties. You will prove from scratch a more general result in an ASMT.

Corollary. (*Properties of $\pi(\cdot)$*). *On P , $\pi(\cdot)$ is*

- *homogeneous of degree 1; and*
- *convex.*

□

Let $y^*(p)$ be the set of solutions to (1). If $\pi(\cdot)$ is differentiable at a point $p \in P$, then $\partial\pi(p)/\partial p_i = y_i^*(p)$ for $i = 1, \dots, L$ (Hotelling's Lemma), implying $y^*(p)$ is single-valued.¹ The next result follows from the Theorem and Hotelling's Lemma.

Corollary 1. (*Properties of $y^*(\cdot)$*). *On P*

- *$y^*(\cdot)$ is homogeneous of degree 0;*
- *If π is C^2 (implying $y^*(\cdot)$ is single-valued and C^1), then the matrix*

$$D^2\pi(p) = \begin{bmatrix} \partial y_1^*/\partial p_1 & \dots & \partial y_1^*/\partial p_L \\ \vdots & & \vdots \\ \partial y_L^*/\partial p_1 & \dots & \partial y_L^*/\partial p_L \end{bmatrix}$$

is a symmetric, positive semidefinite matrix for every $p \in P$. In particular, $\partial y_i^/\partial p_i \geq 0$ for $i = 1, \dots, L$; and $\partial y_i^*/\partial p_j = \partial y_j^*/\partial p_i$ for every i and j between 1 and L .*

¹Hotelling's Lemma, as I call it, of course follows from Theorem 1 in Milgrom and Segal (2002): "Envelope theorems for arbitrary choice sets." *Econometrica* 70.2 (2002): 583-601. This is our second Envelope Theorem from the notes on that topic.